

Remaining schematic work

PetSafe 100-1339 | v0.9.34 | 2026-10-09

11 active issues. Completed steps are archived, not bench requests. S1 electrical contacts are verified; E05 is closed. Q3 stays fitted with identification deferred. Next: LED1 colour checks B42/B43 at docs/FINISHING_MEASUREMENTS.html.

Drawing / connectivity	Remaining data	ERC
0 fitted open pads; 0 open PIC pads; 0 dangling wire ends. 83 native/model net partitions match.	44 ceramic values; LED1 footprint blank. Three local-only PIC branches remain explicit; unsupported R33 withdrawn.	1 retained U6/U6A output conflict. Five stock power-source flags retained; no new copper joins.

Audit boundary

Native wire geometry, library definitions, all catalog pad assignments and existing evidence were checked. Zero open pads does not establish hidden copper or functional behavior. U6A is DNP; its regulator symbol and visible conflict remain by user choice. No new NC markers, net merges or ERC exclusions.

Order of work

Resolve hidden paths and device behavior: S1 contacts are verified and removed from active work. Q3 stays fitted with identification deferred. Three local-only PIC branches, antenna continuations and powered behavior remain. Agent reviews hidden-path candidates from photos; remaining powered checks are staged. The current bench task is LED1 under interfaces below.

Recover critical component values: 44 ceramic values; prioritize tuning and receiver after topology. Use the LCR meter with known test conditions and account for parallel components.

Finish interface and package mapping: Next: two LED1 diode tests to identify the red/green resistor branches. LED footprint/internal pad pairing and motor lead order remain; exact physical fit is later PCB qualification. Use B42/B43 on the labelled LED screen; record colour and voltage or OL. Keep the established body dimensions.

Resolve hidden paths and device behavior

E03 - RF supply and Q8 drive

Established: C5/C6 parallel wiring and Q8 ground/output joins are established. B37 P-F = 1 ohm supports Q8.T2/native2 to C6.1 on H_RF_VDD; no fresh baseline or stability statement supplied. SIL3724A and the U2 inverter remain candidates.

Remaining: Candidate gate/source roles, P-channel turn-off margin and RF timing remain unverified. The source-supply continuity question no longer requires a bench check.

Next: Later operating check on the original hardware: establish source/gate voltages and actual gate-to-source drive, then RF timing if suitable equipment is available. Do not repeat P-F continuity.

Close when: Source path and actual gate-to-source drive are supported; C6 value is handled by V01.

E04 - Antenna and tuning-cell topology

Established: Five PIC-to-control-resistor routes and visible Q3/R13/C10 joins are mapped. Seven midpoint-pair OL exclusions retained. Q3 diode batch complete: L->S latest 0.48 V, L->R 2.3 V; reverse directions and R/S both ways OL.

Remaining: The diode pattern does not confirm the drawn PNP B=R/E=L/C=S mapping. In-circuit paths or contact effects prevent device identification; a single 0.48 V path does not prove a MOSFET. Five capacitor midpoints still lack an established antenna continuation.

Next: User chose to leave Q3 fitted. Isolated identification is deferred; no Q3 desoldering or repeated in-circuit diode batch is requested. Retain the disputed candidate while the agent reviews antenna continuations and other evidence.

Close when: One cell has a supported device map and functional antenna path; the other four are checked only where needed.

E06 - PIC branches without an established onward endpoint

Established: PIC2/RA0 reaches C25/V031; PIC14/RC3 reaches TP11; PIC18/RC7 reaches TP17. RF common-drive wiring is now photo-supported.

Remaining: RA0/C25, RC3/TP11 and RC7/TP17 may terminate locally or continue internally. R3/R4/C28-to-RB3 and D3/R42-to-RB4 circuit roles are unverified.

Next: Review matching front/rear via locations before proposing candidate continuity pairs. TP16-to-TP11/TP17 was already rejected; do not repeat it. No generic all-to-all test is requested.

Close when: Each of the three local-only branches has a traced continuation or explicit evidence of intentional termination; functional roles are qualified separately.

E07 - PIR: Q2 behavior and operating supply

Established: J3/U7 ordering, U6 external pads and PIC21/TP16/Q2 output routing are established. U6 pin5 is tied to VIN; pin4 is unused.

Remaining: Q2 terminal functions, signal behavior and actual U6/PIR operating voltage remain unverified.

Next: Later powered check: record U6 output/J3.1 voltage and compare J3.2 raw signal with PIC21/TP16 as PIR changes state. Agent retains the measured connectivity; no repeat connector or NC-pad checks.

Close when: Q2 signal behavior and U6 output are supported by device or operating evidence.

E08 - Receiver and Q7 operating behavior

Established: U3 input returns and R23/R42/C40, C23/R25/R48 and DNP R47/C42 paths are reconciled. VREF supplies U3 pin5.

Remaining: Q7 transistor pin roles/supply switching, receiver gain/filter response and the functional MCU interface are unverified. H_RX_DETECT is only a series junction, not proof of a missing output wire.

Next: Do not invent another detector wire. In one powered session measure VDD, U3 pin8, VREF and U3 outputs pins1/7, idle and active. Interpret waveforms only after critical capacitor values are recovered.

Close when: Receiver power and bias are supported and the existing PIC25/RA1 interfaces have a documented role; no presumed output destination is used as a closure criterion.

E09 - Supply voltages and Q1 control

Established: BATTERY+, post-Q1 VSYS, VDD and VREF are separate nets. L1/L2 supply sides share VSYS. Five ERC source declarations now describe existing supply paths.

Remaining: Actual voltages and Q1 gate behavior have not been measured. Candidate regulator ratings are not measured board voltages.

Next: Later powered session using the original battery: BATTERY+, VSYS, VDD, VREF, U6 output, Q1 gate and switched receiver supply, idle and active. Keep Q8 timing with E03.

Close when: Actual supply levels and control behavior are recorded without merging battery, VSYS or filtered branches.

E10 - D6 breakdown and ICSP compatibility

Established: D6 anode is GND and cathode is VPP; forward drop is about 0.7V and reverse diode mode is OL. J1 pin order is resolved.

Remaining: G3 does not uniquely establish breakdown voltage. The displayed 3.9V candidate is not a measured clamp rating and could conflict with programming voltage.

Next: No more continuity or ordinary diode tests. Agent seeks a matching primary marking specification; if still ambiguous, plan a current-limited isolated breakdown test. Do not apply a high-voltage test through the PIC.

Close when: Clamp class/rating is established or explicitly bounded for reconstruction, and programming compatibility is checked.

I01 - Candidate device identities and pin roles

Established: Manufacturer candidate datasheets and pin tables exist. D2 local wiring is complete; measured R -> S -> L diode behavior supports a series pair.

Remaining: Exact fitted identity/ratings remain candidates for D1-D6, Q1-Q8/Q11, U1/U2/U6. The 4P marking alone does not identify three-pin D2; a manufacturer 4P code for a two-pin diode is incompatible.

Next: Resolve electrical risks through E03/E04/E07/E08/E09/E10. For remaining ambiguity retain the supported equivalent circuit and disclose ratings not established; no sharper-marking request or repeat D2 rail sweep. Exact manufacturer recovery is not always possible with a meter.

Close when: Each part has a supported pin/function model and adequate ratings or an explicit accepted-equivalent limitation; exact branding is not falsely claimed.

Recover critical component values

V01 - Capacitor values and passive ratings

Established: C5 is 470uF/16V. L1/L2 readings are 1.4/2.2uH. Retained marked resistor nominal values are recorded.

Remaining: 44 fitted ceramic values remain unknown. L1/L2 measurement conditions and component ratings are not established.

Next: After tuning topology, prioritize antenna/RF/receiver capacitors with the LCR meter. Record frequency, mode and isolation. Parallel parts give a combined reading; lift one terminal only where an individual value is needed. Report existing L1/L2 measurement conditions without remeasuring yet.

Close when: Critical individual values are recovered; any bypass substitutions and unverified ratings are explicitly identified assumptions.

All unrecovered ceramic values

C1, C2, C6, C7, C8, C9, C10, C11, C12, C13, C14, C15, C16, C17, C18, C19, C20, C21, C22, C23, C24, C25, C26, C27, C28, C29, C30, C31, C32, C33, C34, C35, C36, C37, C38, C39, C40, C41, C43, C44, C45, C46, C47, C48

Finish interface and package mapping

I02 - LED channel map and motor lead orientation

Established: LED1 is red/green in a 1.5 mm square body; two control branches and common ground are mapped. U5 motor outputs are modeled.

Remaining: Which resistor branch lights red/green and exact LED pad numbering remain unknown. Physical motor lead A/B correspondence is provisional.

Next: B42/B43: unpowered diode mode, black K and red A then B on the LED screen. Report colour and voltage/OL. A reaches R31/PIC22; B reaches R30/PIC23 by visible traces. Internal grounded-pad pairing and manufacturer numbering remain separate; motor lead order is reviewed from photos first.

Close when: LED internal channels/polarity and motor connector physical ordering are supported.

M01 - LED footprint and mechanical qualification

Established: LED1 is the only unassigned footprint. C5 diameter/height, S1 body/actuator size, R8 and C11 sizes are already measured.

Remaining: LED1 land pattern/numbering is the only blank footprint. C5 pitch and S1 pad fit/3D heights are reproduction qualifications, not unknown schematic wires.

Next: Obtain LED underside/pad geometry or a matching vendor drawing before selecting a footprint. For later PCB reproduction check C5 lead pitch and S1 pad fit; use actual 16mm capacitor and 6mm switch heights. Do not repeat body dimensions.

Close when: LED footprint has a supported pad map; remaining package-fit assumptions are documented separately from schematic connectivity.

Scope and sources

Main-board schematic finalization. Routed PCB/layer stack, PIR daughterboard internals and firmware reconstruction remain later work, not completed. Empty-option population design is not needed to reproduce the observed DNP board.

DNP options retain all modeled local routes and candidate roles. J6.2/R36.1 terminates at an unpopulated external interface; no additional onboard rail is assumed. U6A stays a regulator symbol with its visible output conflict by explicit user choice.

All previous issue dispositions: evidence/finalization_review_v0931.json. Native geometry and uncertainty coverage: evidence/finalization_audit.json. Current register: evidence/remaining_work.json. Archived observations and old checklists: docs/history/README.md.

Power-source flags describe supply paths through existing passive components; they do not prove voltage or transistor operation. Installed KiCad 10 libraries and project symbols match the embedded pin/graphic definitions. GUI library reload has not been verified.

Reference rails

Rail	Existing modeled path
BATTERY+	Battery positive -> Q1 single pad/native3 (candidate drain); TP14/R43.2 on the raw side.
VSYS	Q1.L/native2 -> U5/MX512H pin4; L1.1 and L2.1 supply sides.
VDD	TP103, PIC20/C32, U5 pin1/VCC, V064/V063/V075/V082. D3.R via V082 is user-confirmed.
VREF	TP104, PIC4/RA2/V035/C39, R21.2/V071. U3.5 and R22 D/left/model2 at 1 ohm.
GND	Battery negative and ground copper.